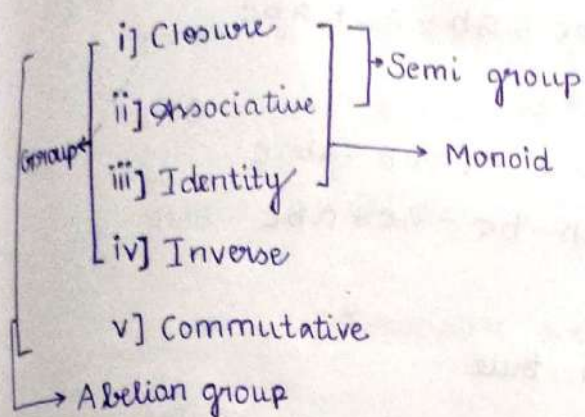




i] Closure property:

$$\forall a, b \in G$$

$$a * b \in G$$

ii] Associative property:

$$\forall a, b, c \in G$$

$$a * (b * c) = (a * b) * c$$

iii] Identity property:

$$\forall a \in G, \exists e \in G$$

such that

$$a * e = e * a = a$$

iv] Inverse

$$\forall a \in G, \exists a^{-1} \in G$$

such that

$$a * a^{-1} = a^{-1} * a = e$$

v] Commutative

$$\forall a, b \in G$$

$$a * b = b * a$$

22-01-25

Problems :

1. Let G be a set of rational nos. except 1 and $*$ defined as $a * b = a + b - ab$, $\forall a, b \in G$, then S.T. $(G, *)$ forms an Abelian group.

Soln. : Proof :

i] Closure $\forall a, b \in G$

$$\text{TPT, } a * b \in G$$

$$a * b = a + b - ab$$

$\therefore$ Closure axiom is true.

ii] Associative $\forall a, b, c \in G$

$$\text{TPT, } a * (b * c) = (a * b) * c$$

$$\begin{aligned} \text{LHS } a * (b * c) &= a * (b + c - bc) \\ &= (a + b + c - bc) - a(b + c - bc) \\ &= a + b + c - bc - ab - ac + abc \end{aligned}$$

$$\begin{aligned} \text{RHS } (a * b) * c &= (a + b - ab) * c \\ &= (a + b - ab) + c - (a + b - ab)c \\ &= a + b + c - ab - bc - ac + abc \end{aligned}$$

$$\text{LHS} = \text{RHS}$$

$\therefore$ Associative axiom is true.

iii] Identity $\forall a \in G, \exists e \in G$

$$\text{TPT, } a * e = e * a = a$$

$$a * e = a$$

$$a + e - ae = a$$

$$e - ae = 0$$

$$e(1 - a) = 0$$

$$e = 0 \in G$$

$\therefore$ Identity axiom is true.

iv] Inverse $\forall a \in G, \exists a^{-1} \in G$

$$a * a^{-1} = a^{-1} * a = e = 0$$

$$a * a^{-1} = 0$$

$$a + a^{-1} - aa^{-1} = 0$$

$$a^{-1}(1 - a) = -a$$

$$a^{-1} = \frac{-a}{1-a} \quad \frac{a}{a-1} \in G \quad (a \neq 1)$$

v] Commutative $\forall a, b \in G$

$$\text{TPT, } a * b = b * a$$

$$a * b = a + b - ab$$

$$b * a = b + a - ba$$

$\therefore$ Commutative axiom is true

shown that

Hence $(G, *)$ forms an abelian group.

2. Let G be a set of positive rational nos. then S.T. $(G, *)$ forms an abelian group.

Soln.: Proof:

i] Closure $\forall a, b \in G$

$$\text{TPT } a * b \in G$$

$$a * b = \frac{ab}{3} \in G$$

Closure axiom is true.

ii] Associative $\forall a, b, c \in G$

$$\text{TPT, } a * (b * c) = (a * b) * c$$

$$\text{LHS } a * (b * c) = a * \frac{bc}{3} = \frac{abc}{9}$$

$$\text{RHS } (a * b) * c = \frac{ab}{3} * c = \frac{abc}{9}$$

$\therefore$ Associative axiom is true.

iii] Identity

$$a * e = e * a = a$$

$$a * e = a$$

$$\frac{ae}{3} = a$$

$$e = 3 \in G$$

$\therefore$ Identity axiom is true.

iv] Inverse $\forall a \in G, \exists a^{-1} \in G$

$$\text{TPT, } a * a^{-1} = a^{-1} * a = e = 3$$

$$a * a^{-1} = 3 \Rightarrow \frac{aa^{-1}}{3} = 3$$

$$\Rightarrow a^{-1} = \frac{9}{a}$$

Inverse axiom is true.

v] Commutative $\forall a, b \in G$

$$a * b = b * a$$

$$a * b = \frac{ab}{3} = \frac{ba}{3} = b * a$$

Hence, $(G, *)$ is true.

$\therefore$ Commutative axiom is true.

23.01.25

3. If M_2 is the set of all 2×2 non-singular matrices over $\mathbb{R}$,

$$M_2 = \left\{ \begin{bmatrix} a & b \\ c & d \end{bmatrix} / a, b, c, d \in \mathbb{R} \text{ and } ad - bc \neq 0 \right\}. \text{ PT } M_2 \text{ forms}$$

a group under matrix multiplication. Is it an abelian?

Soln.: i] Closure $\forall A, B \in M_2$

$$\text{TPT, } AB \in M_2$$

Closure axiom is true.

Matrix multiplication is closed

$$A = \begin{bmatrix} a_1 & b_1 \\ c_1 & d_1 \end{bmatrix}, B = \begin{bmatrix} a_2 & b_2 \\ c_2 & d_2 \end{bmatrix} \in M_2$$

$$AB = \begin{bmatrix} a_1a_2 + b_1c_2 & a_1b_2 + b_1d_2 \\ a_2c_1 + c_2d_1 & b_2c_1 + d_1d_2 \end{bmatrix} \in M_2$$

ii] Associative $\forall A, B, C \in M_2$

TPT, $A(BC) = (AB)C$

Matrix multiplication is always associative.

iii] Identity $\forall A \in M_2, \exists I \in M_2$

TPT, $AI = IA = A$

$\therefore$ Identity elt. $= I = I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \in M_2$

$\therefore$ Identity axiom is true.

iv] Inverse $\forall A \in M_2, \exists A^{-1} \in M_2$ s.t.

$AA^{-1} = A^{-1}A = I$

$\therefore$ Inverse axiom is true

Inverse elt. $= A^{-1} = \frac{1}{|A|} \text{adj} A \quad |A| \neq 0$

Hence, M_2 forms a group under matrix multiplication.

v] Commutative $\forall A, B \in M_2$

TPT, $AB = BA$

WKT, $AB \neq BA$ in matrix

Hence, commutative axiom is not true.

$\therefore$ It is not an abelian.

4. P.T. $(Z_6, \times_6)$ forms a monoid. Also, find which elements of monoid are invertible?

Soln.: $Z_6 = \{ [0], [1], [2], [3], [4], [5] \}$

Inside fully remainders

Z_6	[0]	[1]	[2]	[3]	[4]	[5]
[0]	0	0	0	0	0	0
[1]	0	1	2	3	4	5
[2]	0	2	4	0	2	4
[3]	0	3	0	3	0	3
[4]	0	4	2	0	4	2
[5]	0	5	4	3	2	1

i] Closure

By composition table, closure axiom is true.

ii] Associative

By composition table, associative axiom is true.

iii] Identity $\forall a \in Z_6,$

$\exists e \in Z_6$ s.t.

$a \times_6 e = e \times_6 a = a$

Identity elt. $e = [1] \in Z_6$

Identity axiom is true.

iv] Invertible elements are [1], [5].

$[1] \times_6 [1] = e = [1]$

$[5] \times_6 [5] = e = [1]$

Hence, $(Z_6, \times_6)$ forms a monoid.

Hence proved.

Sub semi group : [Only closure]

Let $(S, *)$ be a semi group & $T \subseteq S$. If T is closed under $*$ then $(T, *)$ is called sub semi group.

Sub monoid : [Only closure and identity]

Let $(M, *, e)$ be a monoid & $T \subseteq M$. If T is closed under $*$ & $e \in T$ then $(T, *, e)$ is called sub monoid.

Idempotent element :

$a \in G$, is called an idempotent element if it satisfies, $a * a = a$, where $(G, *)$ is the set.

Theorem I :

For any commutative monoid $(M, *)$, the set of all idempotent elements of M forms a sub monoid.

Proof : Let $(M, *)$ be a commutative monoid

$$M_1 = \{ a \in M / a * a = a \}$$

TPT, M_1 forms a submonoid

i] Closure : $\forall a, b \in M_1$

$$\Rightarrow a * a = a$$

$$\Rightarrow b * b = b$$

TPT, $a * b \in M_1$

$$a * b = (a * b) * (a * b)$$

$$(a * b) * (a * b) = (a * b) * (b * a)$$

$$= (a * a) * (b * b)$$

$$= a * b \in M_1$$

Closure axiom is true.

ii] Identity : $e \in M_1$

$$e * e = e$$

Sub group : [Closure, Identity, Inverse]

If $(G, *)$ is a group & $H \subseteq G$ is called a sub group of G if it satisfies closure under $*$, the ^{identity} element $e \in H (e \in G)$.

For any $a \in G$ then $\exists a^{-1} \in H$.

Sub group

Trivial

Proper

$$(G, *) (e, *)$$

Cyclic group: $[a^n] \quad n \in \mathbb{Z}^+$

A group $(G, *)$ is called a cyclic group if $\exists a \in G$ such that for every $e \in H$ can be expressed as a^n where $n \in \mathbb{Z}^+$.

eg: $G = \{1, -1, -i, i\}$

for varying values of $(i)^n$ where $n \in \mathbb{Z}^+$, the elements of G are generated.
 $i^4, i^8, i^{12} = 1$

Properties:

1. Every cyclic is an abelian group.

Proof: $\forall a, b \in G \quad \left| \begin{array}{l} a = x^m \\ b = x^n \end{array} \right| \begin{array}{l} x \in G \\ m, n \in \mathbb{Z}^+ \end{array}$

TPT, $a * b = b * a$

LHS $a * b = x^m * x^n$

$= x^{m+n}$

$= x^n * x^m$

$b * a = \text{RHS}$

Group Homomorphism:

Let $\langle G, * \rangle$ and $\langle H, \Delta \rangle$ be two groups. A mapping $g: G \rightarrow H$ is called group homomorphism from G to H if any $a, b \in G$.

$g(a * b) = g(a) \Delta g(b)$

Semi group Homomorphism:

Let $\langle S, * \rangle$ and $\langle T, \Delta \rangle$ be any two semi groups. A mapping $f: S \rightarrow T$ is said to be semigroup homomorphism. iff $f(a * b) = f(a) \Delta f(b)$, for all $a, b \in S$.

eg: $f: (\mathbb{N}, +) \rightarrow (\mathbb{Z}^+, \times)$

$f(a+b) = f(a) \times f(b)$ then homomorphism.

07.02.25

Theorem II

Let $(S, *)$, (T, Δ) and $(V, \oplus)$ be semigroups and $g: S \rightarrow T$ and $h: T \rightarrow V$ be semigroup homomorphisms. Then $(h \circ g): S \rightarrow V$ is a semigroup homomorphism from $(S, *)$ to $(V, \oplus)$.
 $\rightarrow$ composition

Proof: TPT, $\text{hoy}: S \rightarrow V$

$$\forall a, b \in S$$

$$\text{hoy}(a * b) = h[g(a * b)]$$

$$= h[g(a) \Delta g(b)]$$

$$= h(g(a)) \oplus h(g(b))$$

$$\text{hoy}(a * b) = \text{hoy}(a) \oplus \text{hoy}(b)$$

Hence proved

Order of an element:

Let $\langle G, * \rangle$ be a group. Let $a \in G$ with $a^n = e$, where n is the least +ve integer satisfies above. Then we say n is the order of G or $O(G) = n$.

eg: $\{1, -1, i, -i\}$

Order of i $(i)^4 = 1$

$$(i)^2 = -1$$

Examples:

1. Find all the subgroups of $\langle \mathbb{Z}_6, + \rangle$.

Soln.: $\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$ (or) $\mathbb{Z}_6 = [0], [1], [2], [3], [4], [5]$

$$\langle 0 \rangle = \{0\}$$

$$\langle 1 \rangle = \{1, 2, 3, 4, 5, 0\}$$

$$\langle 2 \rangle = \{2, 4, 0\}$$

$$\langle 3 \rangle = \{3, 0\}$$

$$\langle 4 \rangle = \{4, 2, 0\}$$

$$\langle 5 \rangle = \{5, 4, 3, 2, 1, 0\}$$

Trivial subgroups: $\langle 0 \rangle, \langle 1 \rangle, \langle 5 \rangle$

Proper subgroups: $\langle 2 \rangle, \langle 3 \rangle$

2. Find all subgroups of $(\mathbb{Z}_{12}, +)$.

$$\mathbb{Z}_{12} = \{0, 1, 2, \dots, 11\}$$

$$\langle 2 \rangle = \{2, 4, 6, 8, 10, 0\}$$

$$\langle 3 \rangle = \{3, 6, 9, 0\}$$

$$\langle 4 \rangle = \{4, 8, 0\}$$

$$\langle 6 \rangle = \{6, 0\}$$

Proper subgroups : $\{\langle 2 \rangle, \langle 3 \rangle, \langle 4 \rangle, \langle 6 \rangle\}$

Trivial subgroups : $\{\langle 0 \rangle, \langle 1 \rangle, \langle 5 \rangle, \langle 7 \rangle, \langle 8 \rangle, \langle 9 \rangle, \langle 10 \rangle,$

Co-Set : $\langle 11 \rangle\}$

For any $a \in G$, the set - $a * H = \{a * h / h \in H\}$ the left coset and the set - $H * a = \{h * a / h \in H\}$ is right coset and H is a subgroup of G .

NOTE : Union of distinct cosets $\equiv G$

Problems :

1. Find all the coset of H , where $G = \{1, \omega, \omega^2, \omega^3\}$ and $H = \{1, \omega^2\}$ under multiplication.

Soln.: $G = \{1, \omega, \omega^2, \omega^3\}$ $H = \{1, \omega^2\}$

$$1.H = \{1, \omega^2\}$$

$$\omega.H = \{\omega, \omega^3\}$$

$$\omega^2.H = \{\omega^2, 1\}$$

$$\omega^3.H = \{\omega^3, \omega\}$$

$$\text{Distinct} = \{1.H, \omega.H\}$$

2. Find the right cosets of $\{[0], [2]\}$ in $\langle \mathbb{Z}_4, +_4 \rangle$.

Soln.: $H = \{0, 2\}$

$$\mathbb{Z}_4 = G = \{0, 1, 2, 3\}$$

$$H +_4 0 = \{0, 2\}$$

$$H +_4 1 = \{1, 3\}$$

$$H +_4 2 = \{2, 0\}$$

$$H +_4 3 = \{3, 1\}$$

$$\text{Distinct} = \{H +_4 0, H +_4 1\}$$

2.02.2025

Lagrange's Theorem :

The order of each subgroup of a finite group is a divisor of the order of the group.

Proof : Let $(G, *)$ be a finite group with n elts.

Let $(H, *)$ be a subgroup of G with m elts.

$$\text{TPT, } \frac{O(G)}{O(H)} = \frac{n}{m}$$

$$\text{Let } H = \{h_1, h_2, \dots, h_m\}$$

$$\text{Right coset } H * a = \{h_1 * a, h_2 * a, \dots, h_m * a\}, a \in G$$

WKT, The union of all distinct cosets will be equal to G .

$$G = (H * a_1) \cup (H * a_2) \cup \dots \cup (H * a_k)$$

$$O(G) = O(H * a_1) + O(H * a_2) + \dots + O(H * a_k)$$

$$\Rightarrow n = m + m + \dots + m$$

$$n = km$$

$$k = \frac{n}{m}$$

Hence proved

Kernel of Homomorphism :

Let $F: G \rightarrow G$ be a group homomorphism. The set of all elements of G which are mapped into e , identity of G , is called kernel of F .

$$\ker(F) = \{a \in G / F(a) = e\}$$

Theorem IV

The necessary and sufficient condition for a non-empty subset H of a group $(G, *)$ to be a subgroup is $a, b \in H$ then $a * b^{-1} \in H$.

H is a subgroup $\Leftrightarrow a, b \in H$ then $a * b^{-1} \in H$

Proof: Let H be a subgroup of G .

$$\forall a, b \in H$$

$$a, b^{-1} \in H \Rightarrow a * b^{-1} \in H$$

$$\text{Let } a, b \in H \Rightarrow a * b^{-1} \in H$$

TPT, H is a subgroup of G .

Identity: Let $a \in H, b \in H$

$$\Rightarrow a * b^{-1} \in H$$

if $b = a$,

$$a * a^{-1} \in H$$

$$e \in H$$

$\therefore$ Identity element exists.

Inverse: Let $e \in H, a \in H$

$$\Rightarrow e * a^{-1} \in H$$

$$\Rightarrow a^{-1} \in H$$

$\therefore$ Inverse element exists.

Closure: Let $a \in H, b \in H$

$$a, b^{-1} \in H$$

$$\Rightarrow a * (b^{-1})^{-1} \in H$$

$$\Rightarrow a * b \in H$$

$\therefore$ Closure axiom is true.

Hence H is a subgroup of G .

Group code / Coding Theory :

If $B = \{0, 1\}$ then $B^n = \{x_1, x_2, x_3, \dots, x_n \mid x_i \in B, i = 1, 2, \dots\}$ is a group under the binary operation of addition modulo 2 ($\oplus$), then $(B^n, \oplus)$ is called group code.

Hamming distance :

Suppose if x and y represent binary strings $x_1 x_2 x_3 \dots x_n$ and $y_1 y_2 y_3 \dots y_n$ then the no. of positions in the strings for which $x_i \neq y_i$ is called Hamming distance.

Eg: $x = 1011$ $HD(x, y) = 1 + 1 + 1 + 1 = 4$
 $y = 0100$ $HD(y, z) = 1 + 1 + 1 = 3$
 $ry = 1010$ $HD(x, ry) = 1$

Weight :

The number of ones in the binary string $x \in B^n$ is called the weight of x & is denoted by $|x|$ or $w(x)$.

Eg: $x = 1011 \rightarrow w(x) = 3$
 $y = 0100 \rightarrow w(y) = 1$
 $ry = 1010 \rightarrow w(ry) = 2$

Minimum distance (or) minimum hamming distance :

The minimum distance of a code is defined as the minimum of hamming distances btw. all pairs of encoded words in that code.

$x = 1011$ $HD(x, y) = 4$ $HD(y, z) = 3$

$y = 0100$ $HD(x, ry) = 1$

$ry = 1010$ Here, the minimum distance is 1.
($HD(x, z) = 1$)

Encoding function :

E is a mapping, $E : B^m \rightarrow B^n$, where m is the message & n is the encoded message, where $m < n$ always.

Generator matrix (G):

$G = [I_m / A]$, where I_m is the identity matrix and A is chosen random matrix of dimension $m \times (n-m)$.

Parity Check Matrix:

$$H = [A^T / I_{n-m}]$$

18.02.2025

Note: If w is a message $\in B^m$, then $E(w) = wG$ and the code (the set of code words) $C = E(B^m) \subseteq B^n$, where w is a $(1 \times m)$ vector.

Examples:

1. Find the code words generated by the encoding function

$E: B^2 \rightarrow B^5$ w.r.t parity check matrix.

Soln.: $B^m \rightarrow B^n$

$$E(w) = wG \quad B^2 = \{00, 01, 10, 11\}$$

$$H = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$I_{5-2} = I_3 \quad H = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 1 \end{bmatrix} \Rightarrow H = [A^T / I_{n-m}]$$

$$G = [I_m / A] = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

first fill A

$$E(00) = \begin{bmatrix} 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0+0 & 0+0 & 0+0 & 0+0 & 0+0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$E(01) = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0+0 & 0+1 & 0+0 & 0+1 & 0+1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & 1 & 0 \end{bmatrix}$$

$$E(10) = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1+0 & 0+0 & 0+0 & 1+1 & 1+0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$E(11) = \begin{bmatrix} 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 1+0 & 0+1 & 0+0 & 1+1 & 1+1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 \end{bmatrix}$$

2. Find the code words generated by the parity check matrix

$$H = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad \text{when the encoding function } E: \mathbb{B}^3 \rightarrow \mathbb{B}^6$$

Soln.:

$$E(w) = wG$$

$$\mathbb{B}^3 = \{000, 100, 010, 001, 110, 101, 011, 111\}$$

$$H = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 1 \end{bmatrix} \Rightarrow H = A^T / I_{n-m}$$

$$G = [I_m / A] = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

$$E(000) = [0 \ 0 \ 0] \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

$$= [0+0+0 \ 0+0+0 \ 0+0+0 \ 0+0+0 \ 0+0+0 \ 0+0+0]$$

$$E(100) = [1 \ 0 \ 0] \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

$$= [1+0+0 \ 0+0+0 \ 0+0+0 \ 1+0+0 \ 1+0+0 \ 1+0+0]$$

$$= [1 \ 0 \ 0 \ 1 \ 1 \ 1]$$

$$E(010) = [0 \ 1 \ 0] \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

$$= [0+0+0 \ 0+1+0 \ 0+0+0 \ 0+1+0 \ 0+0+0 \ 0+1+0]$$

$$= [0 \ 1 \ 0 \ 1 \ 0 \ 1]$$

$$E(001) = [0 \ 0 \ 1] \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

$$= [0+0+0 \quad 0+0+0 \quad 0+0+1 \quad 0+0+0 \quad 0+0+1 \quad 0+0+1]$$

$$= [0 \quad 0 \quad 1 \quad 0 \quad 1 \quad 1]$$

$$E(110) = [1 \ 1 \ 0] \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

$$= [1+0+0 \quad 0+1+0 \quad 0+0+0 \quad 1+1+0 \quad 1+0+0 \quad 1+1+0]$$

$$= [1 \ 1 \ 0 \ 0 \ 1 \ 0]$$

$$E(101) = [1 \ 0 \ 1] \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

$$= [1+0+0 \quad 0+0+0 \quad 0+0+1 \quad 1+0+1 \quad 1+0+1 \quad 1+0+1]$$

$$= [1 \ 0 \ 1 \ 0 \ 0 \ 0]$$

$$E(011) = [0 \ 1 \ 1] \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

$$= [0+0+0 \quad 0+1+0 \quad 0+0+1 \quad 0+1+0 \quad 0+0+1 \quad 0+1+1]$$

$$= [0 \ 1 \ 1 \ 1 \ 1 \ 0]$$

$$E(111) = [1 \ 1 \ 1] \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

$$= [1+0+0 \quad 0+1+0 \quad 0+0+1 \quad 1+1+0 \quad 1+0+1 \quad 1+1+1]$$

$$= [1 \ 1 \ 1 \ 0 \ 0 \ 1]$$

19.02.2025

10M

Error finding in decoder message:

$Hx^T \rightarrow$ to decode:

Note: $Hx^T = [0]$ - messages are decoded correctly

$Hx^T =$ any column of H - coded should be corrected.

$Hx^T =$ neither of above - there is no unique solution.

Examples :

1. Given the generator matrix $G \equiv \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 & 1 \end{bmatrix}$, corresponding to the encoding function $E: B^3 \rightarrow B^6$, find the corresponding parity check matrix and use it to decode the received words and hence, to find the original message. Are all the words decoded uniquely?

i) 1 1 0 1 0 1 ii) 0 0 1 1 1 1 iii) 1 1 0 0 0 1 iv) 1 1 1 1 1 1

If we assume that $G = [I_3 / A]$, then

Soln.: $H = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$

i) $x = [1 \ 1 \ 0 \ 1 \ 0 \ 1]$

$$Hx^T = \begin{bmatrix} 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \\ 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1+0+0+1+0+0 \\ 1+1+0+0+0+0 \\ 0+1+0+0+0+1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

It has ~~an~~ error.

Hence, original message is 110.

ii) $x = [0 \ 0 \ 1 \ 1 \ 1 \ 1]$

$$Hx^T = H \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0+0+1+1+0+0 \\ 0+0+0+0+1+0 \\ 0+0+1+0+0+1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

It has an error.

It matches with 5th column of H.

Hence, the correct decoded message is $[0 \ 0 \ 1 \ 1 \ 0 \ 1]$

Original message is 001.

$$\text{iii) } x = \begin{bmatrix} -1 & 1 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$Hx^T = H \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1+0+0+0+0+0 \\ 1+1+0+0+0+0 \\ 0+1+0+0+0+1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

It has an error.

It matches with 4th column of H.

Correct decoded message is $[1 \ 1 \ 0 \ 1 \ 0 \ 1]$

Hence, original message is 110.

$$\text{iv) } x = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

$$Hx^T = H \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1+0+1+1+0+0 \\ 1+1+0+0+1+0 \\ 0+1+1+0+0+1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

It has an error.

It does not match with any of the columns of H. $\therefore$ There is no unique solution.

2. Find the code words generated by the parity check matrix $H = \begin{bmatrix} 1 & 1 & 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 1 \end{bmatrix}$, when the encoding function is $e: B^3 \rightarrow B^6$.

Soln.: $G = [I_m / A] \quad E(w) = wG$

$$G = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

$B^m \rightarrow B^n$

$$B_3 = \{000, 100, 010, 001, 110, 101, 011, 111\}$$

$$E(000) = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}_{1 \times 3} G_{3 \times 6} = E(000)_{1 \times 6}$$

$$= \begin{bmatrix} 0+0+0 & 0+0+0 & 0+0+0 & 0+0+0 & 0+0+0 & 0+0+0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}_{1 \times 6}$$

$$E(100) = [1 \ 0 \ 0] G$$

$$= [1+0+0 \ 0+0+0 \ 0+0+0 \ 1+0+0 \ 1+0+0 \ 0+0+0]$$

$$= [1 \ 0 \ 0 \ 1 \ 1 \ 0]$$

$$E(010) = [0 \ 1 \ 0] G$$

$$= [0+0+0 \ 0+1+0 \ 0+0+0 \ 0+1+0 \ 0+0+0 \ 0+1+0]$$

$$= [0 \ 1 \ 0 \ 1 \ 0 \ 1]$$

$$E(001) = [0 \ 0 \ 1] G$$

$$= [0+0+0 \ 0+0+0 \ 0+0+1 \ 0+0+0 \ 0+0+1 \ 0+0+1]$$

$$= [0 \ 0 \ 1 \ 0 \ 1 \ 1]$$

$$E(110) = [1 \ 1 \ 0] G$$

$$= [1+0+0 \ 0+1+0 \ 0+0+0 \ 1+1+0 \ 1+0+0 \ 0+1+0]$$

$$= [1 \ 1 \ 0 \ 0 \ 1 \ 1]$$

$$E(101) = [1 \ 0 \ 1] G$$

$$= [1+0+0 \ 0+0+0 \ 0+0+1 \ 1+0+0 \ 1+0+1 \ 0+0+1]$$

$$= [1 \ 0 \ 1 \ 1 \ 0 \ 1]$$

$$E(011) = [0 \ 1 \ 1] G$$

$$= [0+0+0 \ 0+1+0 \ 0+0+1 \ 0+1+0 \ 0+0+1 \ 0+1+1]$$

$$= [0 \ 1 \ 0 \ 1 \ 1 \ 0]$$

$$E(111) = [1 \ 1 \ 1] G$$

$$= [1+0+0 \ 0+1+0 \ 0+0+1 \ 1+1+0 \ 1+0+1 \ 0+1+1]$$

$$= [1 \ 1 \ 1 \ 0 \ 0 \ 0]$$

29. Given the generator matrix $G = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}$, corresponding to the encoding function $e: \mathbb{B}^4 \rightarrow \mathbb{B}^7$, find the corresponding parity check matrix

and use it to decode the following received words & hence, to find the original message :

1 1 0 0 0 0 1, 1 1 1 0 1 1 1, 0 0 1 0 0 0 1, 0 0 1 1 1 0 0

Soln.:

$$H = \begin{bmatrix} 1 & 1 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 0 & 0 & 1 \end{bmatrix} \quad H = [A^T | I_{n-m}]$$

I_{7-4}
 I_3

i) $x = [1 \ 1 \ 0 \ 0 \ 0 \ 0 \ 1]$

$$Hx^T = H \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1+1+0+0+0+0+0 \\ 1+0+0+0+0+0+0 \\ 0+1+0+0+0+0+1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

It has an error.
It matches with 6th column of H.

Correct decoded message is $[1 \ 1 \ 0 \ 0 \ 0 \ 1 \ 1]$.

Hence, original message is 1 1 0 0.

ii) $x = [1 \ 1 \ 1 \ 0 \ 1 \ 1 \ 1]$

$$Hx^T = H \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \\ 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1+1+0+0+1+0+0 \\ 1+0+1+0+0+1+0 \\ 0+1+1+0+0+0+1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

It has an error.
It matches with the 4th column of H.
Correct decoded message is $[1 \ 1 \ 1 \ 1 \ 1 \ 1 \ 1]$.

Hence, original message is 1 1 1 1.

iii) $x = [0 \ 0 \ 1 \ 0 \ 0 \ 0 \ 1]$

$$Hx^T = H \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0+0+0+0+0+0+0 \\ 0+0+1+0+0+0+0 \\ 0+0+1+0+0+0+1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$$

It has an error.

It matches with 6th column of H.

Correct decoded message is $[0 \ 0 \ 1 \ 0 \ 0 \ 1 \ 1]$

Hence, original message is 0010.

iv) $g_1 = [0 \ 0 \ 1 \ 1 \ 1 \ 0 \ 0]$

$$H g_1^T = H \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \\ 1 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0+0+0+1+1+0+0 \\ 0+0+1+1+0+0+0 \\ 0+0+1+1+0+0+0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

It has no error.

Hence, original message is 0011.